

Tribhuvan University
Institute of Science and Technology
2068

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 113)
(Physics)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Section A:

Attempt any four questions:

1. What are non-inertial frames of reference? Define and explain centrifugal and Coriolis forces. (2-1.2+3)
2. State and explain law of conservation of angular momentum. Also state and explain Kepler's 2nd law. (1+2+1+3)
3. What do you mean by a harmonic oscillator? Discuss the oscillation of diatomic molecule. Hence sketch the energy level diagram. (2+4+1)
4. Discuss the boundary conditions on the field vectors E and D? (3.5+3.5)
5. Explain the meanings of power and power factors. Further discuss the phenomena of resonance and hence obtain quality factor.

Section B:

Attempt any eight questions:

6. A proton is accelerated through a p.d. 50 and then it is allowed to cross a field free region 7.5m long. Find the time required to cross this distance. (4)
7. The initial positions of two particles are (-2, 0) and (0, -2) and they start simultaneously along the axes of x and y with uniform velocities 3i cm/s and 4j cm/s respectively. Obtain the vector representing the position of the 2nd particle with respect to the first as a function of time. (4)
8. Show that the force defined by $F = (-2x^2 + 2xyj)$ is conservative. (4)
9. A particle of mass 5 gm lies in a potential field $V = (8x^2 + 200)$ ergs/gm. Calculate its time period. (4)
10. Calculate the mass of water flowing in 10 minutes through a tube 0.1 cm in diameter 40 cm long, if there is a constant pressure head of 20 cm of water. (11 for water = 0.0089 cgs units). (4)
11. Two small identical conducting spheres have charges of $2.0 \times 10^{-9}C$ and $-0.5 \times 10^{-9}C$, respectively. When they are placed 4 cm apart, what is the force between them? (4)
12. Find the electric field produced by a uniformly polarized sphere of radius R. (4)
13. Find the energy of a uniformly charged spherical shell of total charge q and radius R. (4)
14. A real capacitor C has a parallel leakage resistance R; it is connected in series with an ideal inductance L. Calculate JZI ; find the approximate values at high and low frequencies assuming R is large. (2+2)
15. Calculate the energy density of uniform magnetic field of strength I Tesla in vacuum. ($\mu_0 = 4\pi \times 10^{-7} \text{ NS}^2/\text{C}^2$) (4)